

UTSAV BHARDWAJ

New Delhi, India

+91-7979930500

utsavjha.me@gmail.com

Linkedin

Github

Leetcode

Education

Bharati Vidyapeeth's College of Engineering

B.Tech in Computer Science and Engineering – **SGPA: 9.11**

New Delhi, India

Nov 2022 – June 2026

Work Experience

DRDO (Defence Research and Development Organisation)

Intern – Ongoing

SSPL – New Delhi

- Centralized **25 years of CSV and paper-based R&D project records** from multiple departments into a single structured dataset using Python, enabling consistent reporting across **10 research labs**
- Developed an **end-to-end Power BI dashboard** using **DAX measures** to track **6 KPIs** (budget vs actuals, department-wise overspend, project completion rate, scientist output), providing management a single view to quickly **identify over-budget and stalled projects**

PROJECTS

HiRest – Real-Time Image Sharpening Model | PyTorch, CUDA, Adam Optimizer

[Jun 2025]

- Co-developed **HiRest**, a real-time image sharpening framework using knowledge distillation, designing and training a lightweight CNN (**HINet**) distilled from Restormer that achieved **32.81 dB PSNR**, **0.904 SSIM**, and **30+ FPS** at 720p while reducing the model size to **~3.5M parameters** (**10× smaller** than the teacher model).
- Implemented dual-supervision training combining pixel loss and distillation loss with HIN blocks, residual decoders, CSFF, and SAM modules for computational efficiency and high-quality image restoration.
- Validated model performance using benchmark datasets and MOS evaluation (**4.1/5 vs 4.3/5** teacher), demonstrating transformer-level perceptual quality on consumer GPUs.

Pariksha AI – RAG based Exam Preparation Platform | ChromaDB, API, OCR

[March 2026]

- Developed an **AI-powered exam preparation pipeline** using **ChromaDB**, **Claude API**, and **Tesseract OCR**, where NCERT PDFs are chunked, embedded, and retrieved via **dual RAG queries** to generate CBSE-style questions and evaluate handwritten answers, reducing generation time to under 5 seconds per batch.
- Designed a **vector search system** using **ChromaDB** and **all-MiniLM-L6-v2** embeddings with metadata filters across question type, Bloom's taxonomy level, and chapter, enabling style-calibrated question generation with over 0.80 similarity to real board exam patterns.
- Implemented a **two-stage answer evaluation workflow** combining **Tesseract OCR** for handwritten text extraction and **Claude API** for scoring against RAG-retrieved NCERT key points, delivering structured JSON feedback across 6 rubric dimensions within 8 seconds per submission.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL

AI/ML & GenAI: RAG, Prompt Engineering, LLMs, Claude API, ChromaDB, LangChain, Agentic Workflows

ML Frameworks & Tools: PyTorch, CUDA, Tesseract OCR, REST APIs

Core Subjects: OOPs, Operating Systems, DBMS, Computer Networks, NLP, SDLC

CERTIFICATIONS

- Building with the Claude API By Anthropic
- SQL Intermediate By HackerRank

ACHIEVEMENTS

- Secured **3rd place** in Web Development Hackathon organized by GDSC BVCOE